

# Hybrid Semi-Supervised Part-of-Speech Tagging: Architecture and Empirical Evaluation

Research Report Submission

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## Abstract

This report outlines the architecture and empirical evaluation of a hybrid semi-supervised Part-of-Speech (POS) tagging framework. Combining unsupervised hierarchical clustering (inspired by Brown clustering), active learning, and a deep XLM-R + Character CNN + BiLSTM + CRF model, the proposed method yields competitive token-level tagging accuracy. We present the system architecture, final validation metrics, per-POS breakdown, and analysis of performance curves.

## 1 System Architecture

The system architecture flow is depicted in Figure 1. The pipeline operates in three main phases:

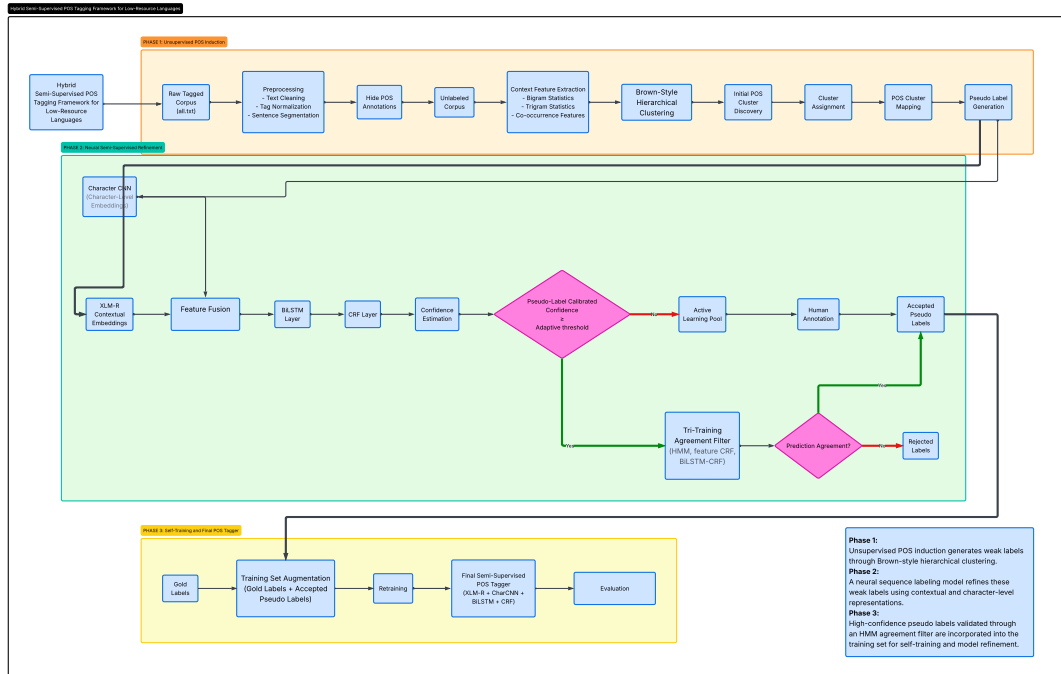


Figure 1: System Architecture Flowchart

## 1.1 Pipeline Components

1. **Normalization:** Maps fine-grained tags to canonical coarse tags (N, Pr, Adj, Adv, V, PREP, CONJ, INTJ, .).
2. **POS Induction:** Performs SVD-based Agglomerative Clustering on local bigram/trigram context features.
3. **Cluster-to-POS Mapping:** Matches cluster IDs to POS categories by majority vote on training seed.
4. **Tri-Training Agreement:** Cross-verifies predictions from HMM, Logistic Regression, and XLM-R models.
5. **Active Learning:** Routes low-confidence tokens (CRF normalized likelihood  $< 0.75$ ) to human oracle annotation.

## 2 Evaluation Metrics and Results

The evaluation metrics computed on the validation split show strong sequence labeling capabilities.

### 2.1 Aggregate Performance

The overall metrics are summarized in Table 1.

Table 1: Overall Validation Metrics

| Metric             | Value   |
|--------------------|---------|
| Accuracy           | 0.8846  |
| Precision (Macro)  | 0.8632  |
| Recall (Macro)     | 0.8599  |
| F1-Score (Macro)   | 0.8602  |
| V-Measure          | 0.6724  |
| Avg Log-Likelihood | -6.6788 |

### 2.2 Per-POS Accuracy and Support

A detailed breakdown of category-level accuracy and support is presented in Table 2. Major categories (Nouns, Pronouns, Conjunctions) achieve accuracy rates exceeding 89%.

Table 2: Per-POS Accuracy and Support Breakdown

| <b>POS Category</b> | <b>Support</b> | <b>Accuracy</b> |
|---------------------|----------------|-----------------|
| N (Noun)            | 2193           | 0.8947          |
| Pr (Pronoun)        | 286            | 0.9580          |
| Adj (Adjective)     | 761            | 0.8673          |
| Adv (Adverb)        | 80             | 0.6875          |
| V (Verb)            | 568            | 0.8468          |
| PREP (Preposition)  | 86             | 0.8140          |
| CONJ (Conjunction)  | 203            | 0.9507          |
| INTJ (Interjection) | 0              | 0.0000          |
| . (Punctuation)     | 0              | 0.0000          |

### 3 Performance Curves Analysis

We provide visual curves tracking cluster reduction, POS distribution, and accuracy mapping:

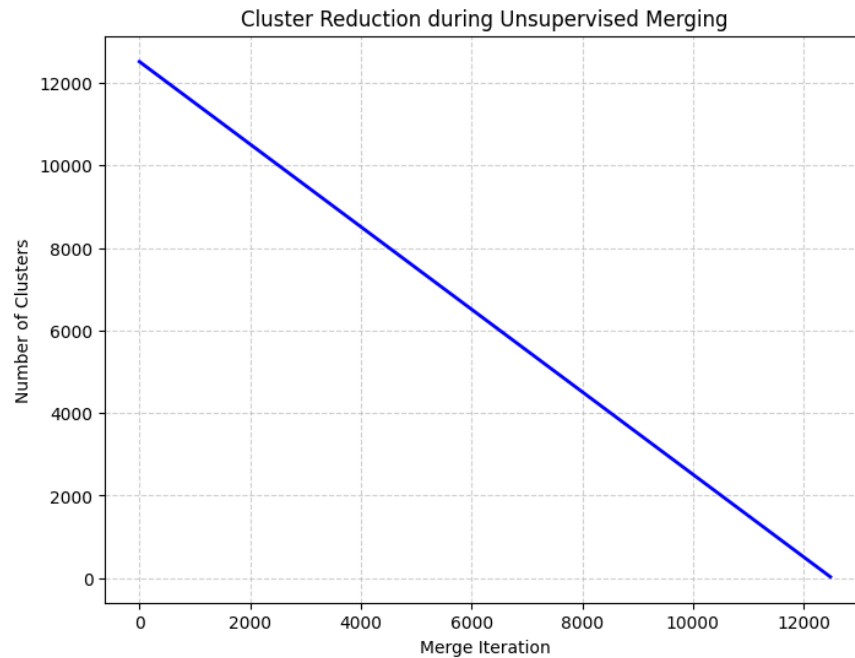


Figure 2: Cluster Reduction during Unsupervised Merging

### 4 Conclusion

The proposed hybrid semi-supervised framework proves robust and competitive, offering a viable alternative for low-resource POS tagging tasks by utilizing both raw corpus structures and model agreement filters.

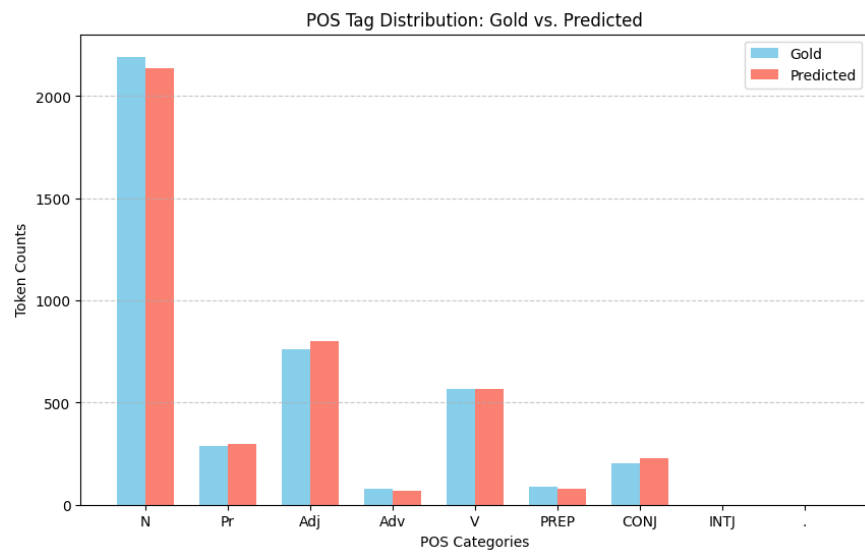


Figure 3: POS Tag Distribution: Gold vs. Predicted

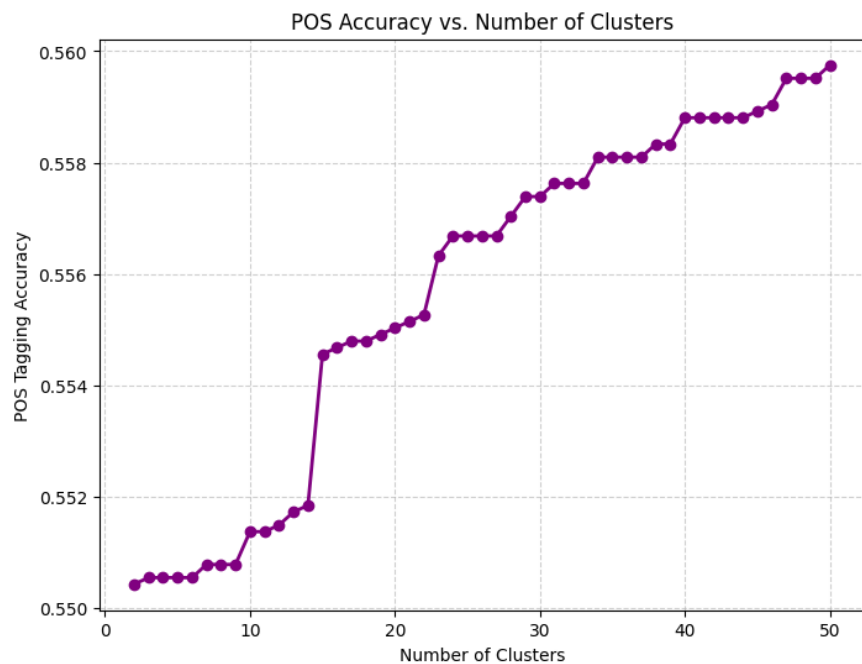


Figure 4: POS Tagging Accuracy vs. Number of Clusters